

Databricks App URL:

<https://ticketing-app-7474643870010414.aws.databricksapps.com/>

Source code Repo URL:

<https://github.com/Abirami29/databricks-lakebase-app-day-1-homework.git>

Screenshot of deployed application:

- Screenshot of V1:

The screenshot displays the 'SyncMetrics Internal Support Dashboard' in a web browser. The dashboard includes a header with navigation icons and a title. Below the header, there are three summary cards: 'Total System Tickets' with a value of 4, 'Open Tickets' with a value of 1, and 'In Progress' with a value of 2. The main content area is divided into two columns. The left column, titled 'Active System Tickets', features a filter dropdown set to 'All' and a list of four tickets. The first ticket, '#4 - Apple watch not syncing [IN_PROGRESS]', is selected. The right column, titled 'Conversation Log for Ticket #4', shows a message from 'Love Apple' stating 'I changed my phone and since the new install of the app the data from apple is not syncing'. Below the conversation log is a section for 'Post an Internal Message Reply' with input fields for 'Author Name/ID' (filled with 'Love Apple') and 'Message Body Text' (filled with the same message as the log). A 'Post Reply Message' button is at the bottom of this section. At the bottom of the dashboard is a section titled '+ Initiate a New Support Ticket' with input fields for 'Short Ticket Title Summary' (filled with 'Apple watch not syncing'), 'Reporter Name/ID' (filled with 'World-is-an-apple'), and 'Initial System Status' (set to 'open').

SyncMetrics Internal Support Dashboard

Total System Tickets: 4 Open Tickets: 1 In Progress: 2

Active System Tickets

Filter Tickets by Current Status: All

Select a Support Ticket to Inspect:

- ☐ #1 - Garmin Sync Data Failing [OPEN]
- ☐ #2 - MyFitnessPal Double Logging Error [IN_PROGRESS]
- ☐ #3 - Vanguard UI Screen Freeze [RESOLVED]
- ☒ #4 - Apple watch not syncing [IN_PROGRESS]

Conversation Log for Ticket #4

Love Apple

I changed my phone and since the new install of the app the data from apple is not syncing

Post an Internal Message Reply

Author Name/ID: Love Apple

Message Body Text: I changed my phone and since the new install of the app the data from apple is not syncing

Post Reply Message

+ Initiate a New Support Ticket

Short Ticket Title Summary: Apple watch not syncing

Reporter Name/ID: World-is-an-apple

Initial System Status: open

ticketing-app-7474643870010414.aws.databricksapps.com

Select a Support Ticket to Inspect:

☐ #1 - Garmin Sync Data Failing [OPEN]

☐ #2 - MyFitnessPal Double Logging Error [IN_PROGRESS]

☐ #3 - Vanguard UI Screen Freeze [RESOLVED]

☒ #4 - Apple watch not syncing [IN_PROGRESS]

+ Initiate a New Support Ticket

Short Ticket Title Summary:

Apple watch not syncing

Reporter Name/ID:

World-is-an-apple

Initial System Status:

open

Submit New Ticket

I changed my phone and since the new install of the app the data from apple is not syncing

Post an Internal Message Reply

Author Name/ID:

Love Apple

Message Body Text:

I changed my phone and since the new install of the app the data from apple is not syncing

Post Reply Message

Modify Ticket Status

Change Status To:

in_progress

Update System Status

- Screenshot of V2 (after improving the visual display of the app):

ticketing-app-7474643870010414.aws.databricksapps.com

SyncMetrics Internal Support Dashboard

Manage and track customer support tickets

Dashboard Overview

Total Tickets

4

Open

1

↑ 25%

In Progress

2

↑ 50%

Resolved

1

↑ 25%

Active System Tickets

Filter by Status

All

Select a Support Ticket:

☒ #1 - Garmin Sync Data Failing

☐ #2 - MyFitnessPal Double Logging Error

☐ #3 - Vanguard UI Screen Freeze

☐ #4 - Apple watch not syncing

OPEN

Conversation Log - Ticket #1

athlete_user_1 · Aug 05, 2026 at 08:13 AM

My watch synced successfully but the JSON payload is empty inside the repository.

support_agent_A · Aug 05, 2026 at 08:13 AM

Investigating. Checking the API endpoint routing logs now.

Add Message

Your Name

Enter your name...

Select a Support Ticket:

- ☒ 📱 #1 - Garmin Sync Data Failing
- ☐ 📱 #2 - MyFitnessPal Double Logging Error
- ☐ 📱 #3 - Vanguard UI Screen Freeze
- ☐ 📱 #4 - Apple watch not syncing

OPEN

+ Create New Ticket

 Ticket Title

Brief description of the issue...

 Reporter

Name or user ID...

 Initial Status

open

🔖 Create Ticket

 support_agent_A - Aug 05, 2026 at 08:13 AM

Investigating. Checking the API endpoint routing logs now.

Add Message

 Your Name

Enter your name...

 Message

Type your message here...

 Send Message

Update Status

Change ticket status to:

 Open

 Update Status

Screenshot of tables created:

The screenshot shows the Databricks SQL Editor interface. The left sidebar contains navigation options: PROJECT (internal-supp...), BRANCH (production), and APP BACKEND (Data API). The main panel displays the 'Tables' section for the 'databricks_postgres' database and 'public' schema. The 'ticket_messages' table is selected, showing its structure with columns: message_id (SERIAL PRIMARY KEY), ticket_id (INTEGER), message_text (TEXT NOT NULL), author (VARCHAR(100) NOT NULL), and created_at (TIMESTAMP DEFAULT CURRENT_TIMESTAMP). The 'CONSTRAINTS' section shows a foreign key constraint 'ticket_messages_ticket_id_fkey' referencing 'public.tickets (ticket_id)' and a primary key constraint 'ticket_messages_pkey' on 'message_id'. The 'INDEXES' section shows a unique index 'ticket_messages_pkey _ USING BTREE (message_id)'.

Table name: ticket_messages, Schema: public, Row Level Security: ☐

COLUMNS Add column

- message_id SERIAL PRIMARY KEY
- ticket_id INTEGER
- message_text TEXT NOT NULL
- author VARCHAR(100) NOT NULL
- created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

CONSTRAINTS Add constraint

- ☐ CONSTRAINT ticket_messages_ticket_id_fkey FOREIGN KEY (ticket_id) REFERENCES public.tickets (ticket_id) ON DELETE CASCADE
- CONSTRAINT ticket_messages_pkey PRIMARY KEY (message_id)

INDEXES Add index

- UNIQUE INDEX ticket_messages_pkey _ USING BTREE (message_id)

POLICIES Add policy

The screenshot shows the Databricks SQL Editor interface. The left sidebar contains navigation options: PROJECT (internal-supp...), BRANCH (production), and APP BACKEND (Data API). The main panel displays the 'Tables' section for the 'databricks_postgres' database and 'public' schema. The 'tickets' table is selected, showing its structure with columns: ticket_id (SERIAL PRIMARY KEY), title (VARCHAR(255) NOT NULL), status (VARCHAR(50) NOT NULL), created_by (VARCHAR(100) NOT NULL), and created_at (TIMESTAMP DEFAULT CURRENT_TIMESTAMP). The 'CONSTRAINTS' section shows a check constraint 'tickets_status_check' and a primary key constraint 'tickets_pkey' on 'ticket_id'. The 'INDEXES' section shows a unique index 'tickets_pkey _ USING BTREE (ticket_id)'.

Table name: tickets, Schema: public, Row Level Security: ☐

COLUMNS Add column

- ticket_id SERIAL PRIMARY KEY
- title VARCHAR(255) NOT NULL
- status VARCHAR(50) NOT NULL
- created_by VARCHAR(100) NOT NULL
- created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

CONSTRAINTS Add constraint

- CONSTRAINT tickets_status_check CHECK (((status)::text = ANY ((ARRAY['open'::character varying, 'in_progress'::character varying, 'resolved'::character varying])::text)))
- CONSTRAINT tickets_pkey PRIMARY KEY (ticket_id)

INDEXES Add index

- UNIQUE INDEX tickets_pkey _ USING BTREE (ticket_id)

POLICIES Add policy

databricks Lakebase Postgres Autoscaling workspace

PROJECT: internal-supp...
 Dashboard
 Branches
 Settings

BRANCH: production
 Genie
 Overview
 Monitoring
 SQL Editor
Tables
 Backup & Restore
 APP BACKEND
 Data API

Projects > internal-support-db > production

Tables
 databricks_postgres
 Schema
 public
 Search...
 ticket_messages
tickets

153ms < 1 - 4 of 4 > ...

ticket_id serial	title varchar(255)	status varchar(50)	created_by varchar(100)	created_at timestamp	ticket_messages
1	Garmin Sync Data Fai...	open	athlete_user_1	2026-08-05 08:13:00.23...	ticket_messages
2	MyFitnessPal Double Lo...	in_progress	athlete_user_2	2026-08-05 08:13:00.23...	ticket_messages
3	Vanguard UI Screen Fre...	resolved	athlete_user_3	2026-08-05 08:13:00.23...	ticket_messages
4	Apple watch not syncing	in_progress	World-is-an-apple	2026-08-05 23:21:11.66...	ticket_messages

ticket_messages

message_id serial	ticket_id integer	message_text text	author varchar(100)	created_at timestamp
7	4	I changed my phone and since...	Love Apple	2026-08-05 23:22:15.001403

Open in sub view

dbc-075872da-bdef.cloud.databricks.com/lakebase/projects/9d0d7691-67b0-4912-8171-399b4daad408/branches/br-muddy-heart-d8buqy6/tables

databricks Lakebase Postgres Autoscaling workspace

PROJECT: internal-supp...
 Dashboard
 Branches
 Settings

BRANCH: production
 Genie
 Overview
 Monitoring
 SQL Editor
Tables
 Backup & Restore
 APP BACKEND
 Data API

Projects > internal-support-db > production

Tables
 databricks_postgres
 Schema
 public
 Search...
 ticket_messages
tickets

152ms < 1 - 7 of 7 > ...

message_id serial	ticket_id integer	message_text text	author varchar(100)	created_at timestamp	ticket
1	1	My watch synced succes...	athlete_user_1	2026-08-05 08:13:00.36...	ticket
2	1	Investigating. Checkin...	support_agent_A	2026-08-05 08:13:00.36...	ticket
3	2	The app is multiplying...	athlete_user_2	2026-08-05 08:13:00.49...	ticket
4	2	We identified a duplic...	support_agent_B	2026-08-05 08:13:00.49...	ticket
5	3	When I click my portfo...	athlete_user_3	2026-08-05 08:13:00.62...	ticket
6	3	Database index optimiz...	support_agent_A	2026-08-05 08:13:00.62...	ticket
7	4	I changed my phone and...	Love Apple	2026-08-05 23:22:15.00...	ticket

Short reflection Q&A:

1. What was the most difficult part?

Vibe coding and troubleshooting with Genie. It kept going on a loop at a point undoing and redoing the same thing over and over again. Need to manually debug the issue and troubleshoot. Unfamiliarity with the platform and locating different things were difficult. But good learning

2. How is Lakebase different from storing this data in a traditional analytics table?

Traditional analytics table wouldn't have referencial index. Lakebase tables are so

similar to OLTP tables designed to handle high frequency concurrent read-writes. Lakebase uses row-based storage optimized for fast point lookups whereas Analytical tables use columnar storage optimized for heavy scanning and aggregations. Lakebase delivers low latency responses for users and AI agents.

3. What feature would you add next?

The next milestone feature would be linking an asynchronous Spark processing script to query the vector search endpoints inside Databricks Mosaic AI, allowing an autonomous AI agent to read ticket histories and auto-generate customer responses.